THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 10 · SET 01

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. A leaf of a potted plant is kept in the dark for 24 hours. A part of it is then covered with black paper, the plant is kept in sunlight for some hours and the leaf is tested with iodine solution. Only the uncovered part turns blue-black. The experiment shows that:

A. chlorophyll is necessary for photosynthesis
B. light is necessary for photosynthesis
C. carbon dioxide is necessary for photosynthesis
D. oxygen is released in photosynthesis

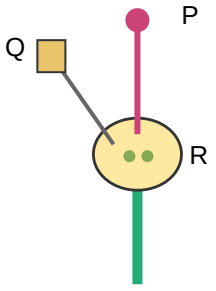
Ans: B Both parts have chlorophyll and CO_2; only the covered part lacks light and shows no starch.

2. A person touches a hot plate and pulls his hand away before he feels the pain. The correct path of the impulse in this action is:

A. receptor → sensory neuron → spinal cord → motor neuron → effector
B. receptor → motor neuron → spinal cord → sensory neuron → effector
C. receptor → sensory neuron → brain → motor neuron → effector
D. effector → sensory neuron → spinal cord → motor neuron → receptor

Ans: A A reflex arc passes through the spinal cord without waiting for the brain.

3. The figure shows a longitudinal section of a flower. Pollen grains germinate on the part labelled P. The male gamete finally fuses with the female gamete present in the part labelled:



Longitudinal section of a flower (simplified)

A. P B. Q C. R D. None of these

Ans: C P is the stigma, Q the anther and R the ovary. The pollen tube carries the male gamete down through the style to the ovule inside the ovary.

4. A pea plant with round, yellow seeds (RrYy) is self-pollinated. The fraction of the offspring expected to have wrinkled, green seeds is:

A. 9/16 B. 3/16 C. 1/16 D. 4/16

Ans: C In the 9 : 3 : 3 : 1 ratio only rryy (wrinkled, green) is 1/16.

5. In a food chain grass → grasshopper → frog → snake → eagle, 10 000 J of energy is available at the grass level. According to the 10% law, the energy available to the eagle is:

A. 1 J B. 10 J C. 100 J D. 1000 J

Ans: A 10 000 → 1000 → 100 → 10 → 1 J after four transfers.

6. Transpiration in plants is useful because it:

A. makes the plant lose all its water so that it can absorb fresh water

B. creates a pull that helps in upward movement of water and minerals and also helps in cooling the plant

C. transports the food made in the leaves to the roots

D. closes the stomata and stops photosynthesis

> **Ans: B** Evaporation of water from leaves creates a suction pull in the xylem and cools the plant.

7. A school compost pit receives (i) vegetable peels, (ii) polythene bags, (iii) dry leaves and (iv) broken glass. Which of these will be broken down by decomposers?

A. (i) and (iii) only

B. (i), (ii) and (iii)

C. (ii) and (iv) only

D. all four

> **Ans: A** Vegetable peels and dry leaves are biodegradable; polythene and glass are not.

8. **Assertion (A):** The sex of a child is determined by the father and not by the mother.
Reason (R): The father produces two types of gametes, carrying X and Y chromosomes, whereas the mother produces only X-type gametes.

> **Ans: A** Egg (X) + sperm (X) → girl; egg (X) + sperm (Y) → boy.

9. **Assertion (A):** Sexual reproduction gives rise to variation among the offspring.
Reason (R): In sexual reproduction the offspring receive a different combination of genes from two parents and errors may occur during DNA copying.

> **Ans: A** Gene combinations from two parents and copying errors are the sources of variation.

10. Why is the rate of breathing in aquatic organisms much faster than in terrestrial organisms?

> The amount of dissolved oxygen in water is much less than the oxygen in air (1), so aquatic organisms must take in more water, i.e. breathe faster, to get enough oxygen (1).

11. State any two advantages of vegetative propagation.

> Plants bearing flowers/fruits can be grown in a short time; seedless plants (banana, orange) can be propagated; the offspring are genetically identical to the parent (any two, 1 each).

O R

Write any two differences between sexual and asexual reproduction.

> Sexual: two parents/gametes, offspring show variation; Asexual: single parent, offspring identical (any two points, 1 each).

12. Why is the use of iodised salt recommended in our diet? Name the hormone whose synthesis needs iodine and the disease caused by its deficiency.

> Iodine is needed by the thyroid gland to make thyroxine (1); its deficiency causes goitre (swollen neck) (1).

13. Name the hormone secreted by the pancreas that controls blood sugar. What will happen if its secretion decreases? State one function of adrenaline.

> Insulin (1). Blood sugar level rises (diabetes) (1). Adrenaline prepares the body for ‘fight or flight’: increases heart rate and breathing, diverts blood to muscles (1).

14. A tall pea plant (TT) is crossed with a dwarf pea plant (tt). Show the cross up to the F_2 generation using a Punnett square and write the phenotypic and genotypic ratios of F_2.

> F_1: all Tt (tall) (½). F_2 from Tt × Tt: TT, Tt, Tt, tt in a Punnett square (1). Phenotypic ratio 3 tall : 1 dwarf (1); genotypic ratio 1 TT : 2 Tt : 1 tt (½).

15. A town produces large quantities of garbage every day: kitchen waste, plastic bottles, paper and old refrigerators. The municipal body wants to segregate the waste. Scientists also warn that the use of certain chemicals in refrigerators and aerosols has thinned the protective ozone layer high in the atmosphere.

(a) What is meant by a biodegradable substance? Name one such item from the waste listed above.

> A substance that can be broken down by biological agents such as bacteria/decomposers (½); kitchen waste or paper (½).

(b) Why are plastic bottles harmful for the environment?

> They are non-biodegradable, remain in the environment for ages, choke drains and harm animals (1).

(c) Name the chemical responsible for ozone depletion and state one harmful effect of the depletion on human health.

> Chlorofluorocarbons (CFCs) (1); more UV radiation reaches the earth causing skin cancer/cataract (1).

OR

(d) In a food chain, why does the number of organisms decrease at higher trophic levels? State the 10% law.

> Only about 10% of the energy is passed to the next trophic level, the rest being used up/lost as heat, so fewer organisms can be supported at the higher levels (2).

16. (a) Describe the main events of sexual reproduction in a human female: name the organ that produces the egg, the site of fertilisation, the site of implantation and the role of the placenta. (b) Name any two methods of contraception and state one disadvantage of each type of method you name.

> (a) Ovary produces eggs (½); fertilisation in the fallopian tube (oviduct) (1); the zygote divides and the embryo is implanted in the uterine wall (1); the placenta provides nutrition and oxygen to the embryo and removes waste (1). (b) e.g. condom (barrier – may tear/fail) (1), oral pills (hormonal – may cause side effects) (1); any two, with a correct disadvantage (½ + ½ each).

OR

(i) Describe how carbohydrates, proteins and fats are digested in the human alimentary canal, naming the enzyme/juice at each stage. (ii) State the role of villi in the small intestine.

> (i) Mouth: salivary amylase converts starch to sugar (1); stomach: HCl and pepsin digest proteins (1); small intestine: bile emulsifies fats, pancreatic amylase/trypsin/lipase and intestinal juice complete digestion (1½). (ii) Villi increase the surface area for absorption and have a rich blood supply to carry the absorbed food (1½).

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. A student burns a strip of magnesium ribbon in air and collects the white ash formed. The ash is dissolved in water and the solution is tested with red and blue litmus paper. Which statement is correct?

A. The ash is MgO and it turns red litmus blue

B. The ash is MgO and it turns blue litmus red

C. The ash is $Mg(OH)_2$ and it turns blue litmus red

D. The ash is $MgCO_3$ and it does not change either litmus

> **Ans: A** $2Mg + O_2 \rightarrow 2MgO$; MgO is a basic oxide, its solution ($Mg(OH)_2$) turns red litmus blue.

18. Which of the following reactions is both a combination reaction and exothermic in nature?

A. $CaCO_3 \rightarrow CaO + CO_2$ B. $CaO + H_2O \rightarrow Ca(OH)_2$

C. $2H_2O \rightarrow 2H_2 + O_2$ D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

> **Ans: B** Quicklime and water combine to give slaked lime with the release of heat. (a) and (c) are decompositions, (d) is displacement.

19. Rahul tests four solutions P, Q, R and S with universal indicator and records pH values 2, 7, 10 and 13 respectively. Which solution is most suitable as a mild antacid for heartburn?

A. P B. Q C. R D. S

> **Ans: C** An antacid must be a mild base: R (pH 10). S (pH 13) is strongly basic and harmful; P is acidic; Q is neutral.

20. Four metals W, X, Y and Z were tested. W reacts with cold water; X reacts only with steam; Y reacts with dilute HCl but not with water or steam; Z reacts with none of these. The correct decreasing order of reactivity is:

A. Z > Y > X > W B. W > X > Y > Z C. X > W > Y > Z D. W > Y > X > Z

> **Ans: B** Reactivity is judged by the mildest reagent that reacts: cold water (W) > steam (X) > dilute acid (Y) > no reaction (Z).

21. The number of covalent bonds (shared pairs of electrons) present in one molecule of ethyne, C_2H_2, is:

A. 3 B. 4 C. 5 D. 6

> **Ans: C** H–C≡C–H has 1 + 3 + 1 = 5 shared pairs.

22. **Assertion (A):** Sodium hydrogencarbonate is used in baking powder.
Reason (R): On heating, it releases carbon dioxide, which makes the cake or bread rise and become soft.

> **Ans: A** Both true; $NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ and the CO_2 explains the use.

23. Balance the equation $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$ and name the type of reaction. State one use of this reaction.

> $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$ (1). Displacement reaction (highly exothermic, thermite reaction); used to weld railway tracks/cracked machine parts (1).

24. Dry HCl gas does not change the colour of dry blue litmus paper but hydrochloric acid solution does. Why?

> Acidic behaviour is shown only when H^+ ions are present (1). Dry HCl gas has no H^+(aq); in water HCl ionises to H^+/H_3O^+ which turns blue litmus red (1).

25. Zinc is obtained from its sulphide ore ZnS. Name the steps involved and write the chemical equation for each step.

> Roasting: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ (1½). Reduction with carbon: $ZnO + C \rightarrow Zn + CO$ (1). Final refining by electrolysis (½).

OR

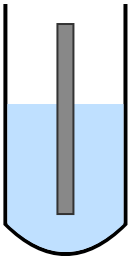
State three physical properties of ionic compounds and give the reason for each.

> High melting/boiling points – strong electrostatic force between ions; soluble in water, insoluble in kerosene; conduct electricity in molten state/solution – ions are free to move, not in solid state (any three, 1 each).

26. Draw the structures of the two isomers of butane (C_4H_{10}) and name them. Also write the IUPAC name of $CH_3CH_2CH_2OH$.

> n-butane $CH_3–CH_2–CH_2–CH_3$ (1); 2-methylpropane (isobutane) $(CH_3)_2CH–CH_3$ (1); propan-1-ol (1).

27. Ramesh puts a few granules of zinc in a test tube containing dilute sulphuric acid. He observes bubbles of a colourless gas and, when a burning matchstick is brought near the mouth of the test tube, the gas burns with a ‘pop’ sound.



Zinc granules in dilute H_2SO_4

(a) Name the gas evolved and state the test confirming it.

> Hydrogen gas; it burns with a pop sound (½ + ½).

(b) Write the balanced chemical equation for the reaction.

> $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow$ (1)

(c) Would the same observation be made if copper turnings were used in place of zinc? Explain using the reactivity series.

> No (½). Copper lies below hydrogen in the reactivity series so it cannot displace hydrogen from dilute acid (1½).

OR

(d) What will you observe if the zinc granules are added to copper sulphate solution? Write the equation.

> Blue colour fades, reddish-brown copper deposits on zinc (1). $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (1).

28. (a) Write the chemical name and formula of bleaching powder. How is it prepared? Write the equation. (b) State two uses. (c) What is observed when carbon dioxide is passed through lime water first in small and then in excess quantity? Write equations.

> (a) Calcium oxychloride, $CaOCl_2$; $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ (2). (b) Bleaching cotton/linen in textile industry; disinfecting drinking water (1). (c) Lime water turns milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$; on passing excess the milkiness disappears: $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ (2).

OR

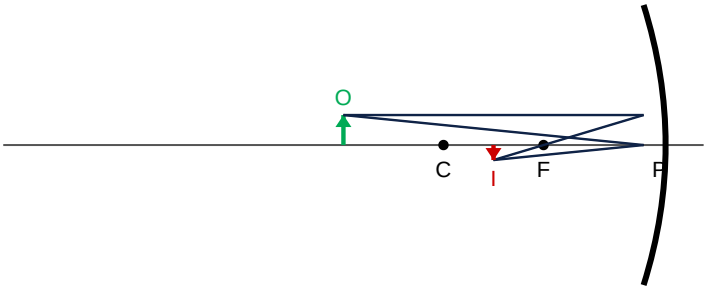
(a) Ethanol reacts with sodium and with ethanoic acid. Write the equation and name the products in each case. (b) How can ethanol be converted into ethene? Write the conditions. (c) Why does ethanol not turn litmus red although it has an –OH group?

> (a) $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$ (sodium ethoxide, hydrogen) (1½); $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ in the presence of conc. H_2SO_4 (ethyl ethanoate, water) (1½). (b) Heating with excess conc. H_2SO_4 at 443 K: $C_2H_5OH \rightarrow CH_2{=}CH_2 + H_2O$ (1). (c) Ethanol does not ionise in water to give H^+, so it is neutral to litmus (1).

SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

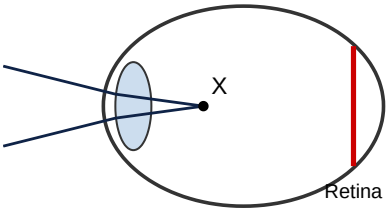
29. A concave mirror of focal length 10 cm forms an image of an object placed 30 cm from it, as shown in the figure. The image is:



A. real, inverted, diminished and between F and C
B. virtual, erect and magnified
C. real, inverted and of the same size at C
D. real, inverted and magnified beyond C

Ans: A u = 30 cm > 2f (20 cm). v = uf/(u − f) = 15 cm, between F (10 cm) and C (20 cm); m = −0.5, so the image is diminished.

30. A student cannot read the writing on the blackboard from the last bench but can read his book comfortably. The defect of vision and the lens needed to correct it are:

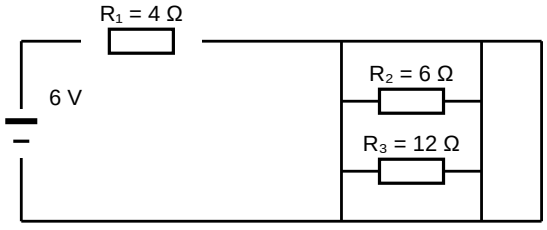


Parallel rays focus at X, in front of the retina

A. myopia; concave lens
B. myopia; convex lens
C. hypermetropia; concave lens
D. hypermetropia; convex lens

Ans: A Near vision is normal, distant vision is blurred: myopia, corrected by a concave lens.

31. Three resistors are connected to a 6 V battery as shown in the figure. The current drawn from the battery is:



A. 0.5 A
B. 0.75 A
C. 1.0 A
D. 1.5 A

Ans: B 6 Ω ∥ 12 Ω = 4 Ω; total R = 4 + 4 = 8 Ω; I = 6/8 = 0.75 A.

32. A straight conductor carries current from left to right in the plane of the paper. A uniform magnetic field acts into the plane of the paper. The force on the conductor is directed:

A. upwards, in the plane of the paper
B. downwards, in the plane of the paper
C. into the plane of the paper
D. out of the plane of the paper

Ans: A Fleming's left-hand rule: forefinger → into the page, middle finger → right, thumb (force) → up.

33. **Assertion (A):** A magnetic compass needle placed near a wire carrying current gets deflected.
Reason (R): A current-carrying conductor produces a magnetic field around it.

> **Ans: A** Oersted's experiment: the field of the wire deflects the needle.

34. A concave mirror has a focal length of 15 cm. An object is placed 10 cm in front of it. Find the position of the image and state its nature.

> u = −10 cm, f = −15 cm; 1/v = 1/f − 1/u = −1/15 + 1/10 = 1/30 → v = +30 cm (1). Image is behind the mirror: virtual, erect and magnified (m = −v/u = +3) (1).

OR

Define the principal focus of a concave mirror. If the radius of curvature of a concave mirror is 24 cm, what is its focal length (with sign)?

> Principal focus: the point on the principal axis at which rays parallel to the axis meet after reflection (1). f = R/2 = −12 cm (1).

35. A myopic person has a far point at 80 cm. Calculate the focal length and power of the lens needed to correct the defect. State the nature of the lens.

> The lens must form a virtual image of a distant object at 80 cm: f = −80 cm = −0.8 m (1). P = 1/f = −1.25 D (1). Concave (diverging) lens (1).

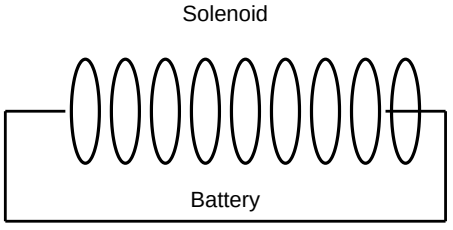
36. Two resistors of 6 Ω and 3 Ω are connected in parallel across a 12 V battery. Calculate (i) the equivalent resistance, (ii) the total current drawn and (iii) the heat produced in the 6 Ω resistor in 10 s.

> (i) 1/R = 1/6 + 1/3 = 1/2 → R = 2 Ω (1). (ii) I = 12/2 = 6 A (1). (iii) Current in 6 Ω = 12/6 = 2 A; H = I^2Rt = 4 × 6 × 10 = 240 J (1).

37. State Snell's law of refraction. The speed of light in a transparent medium is 2×10^8 m/s; find its refractive index (speed of light in air = 3×10^8 m/s). What is the nature of the emergent ray when light passes through a rectangular glass slab?

> sin i / sin r = constant (= refractive index) (1). n = c/v = $3 \times 10^8 / 2 \times 10^8$ = 1.5 (1). The emergent ray is parallel to the incident ray but laterally displaced (1).

38. Ananya makes an electromagnet by winding insulated copper wire on a cardboard tube (a solenoid) and connecting it to a battery. She finds that the solenoid behaves like a bar magnet. She also pushes a bar magnet in and out of a coil connected to a galvanometer and notices that the galvanometer needle deflects.



(a) What is observed in the galvanometer when the magnet is pushed in and pulled out? Name the phenomenon.

> The needle deflects in opposite directions (½); electromagnetic induction (½).

(b) Write any two ways of increasing the strength of the magnetic field of the solenoid.

> Increasing the current, increasing the number of turns, inserting a soft iron core (any two, ½ each).

(c) Draw the pattern of the magnetic field lines of the solenoid. Which end of the solenoid will act as the north pole?

> Field lines inside parallel straight lines, outside like those of a bar magnet (1); the end from which the field lines emerge (where the current appears anticlockwise) is the north pole (1).

OR

(d) State Fleming's right-hand rule. Name a device based on it.

> Stretch thumb, forefinger and middle finger mutually perpendicular: forefinger → magnetic field, thumb → motion of conductor, middle finger → direction of induced current (1½); electric generator (½).

39. An object 5 cm high is placed 30 cm from a convex lens of focal length 20 cm. Find the position, nature and size of the image formed. Draw a ray diagram for the situation.

> u = −30 cm, f = +20 cm. 1/v = 1/f + 1/u = 1/20 − 1/30 = 1/60 → v = +60 cm (1½). m = v/u = −2 (1). h′ = −2 × 5 = −10 cm (½). Image is real, inverted, magnified, 10 cm tall, 60 cm from the lens on the other side (1). Correct ray diagram with two rays and image (1).

OR

(a) With a labelled diagram, explain the principle, construction and working of a simple electric motor.
(b) What is the function of the split ring (commutator)?

> (a) Principle: a current-carrying coil in a magnetic field experiences a force/torque (1). Labelled diagram: rectangular coil ABCD, magnets N–S, split-ring commutator, brushes, battery (1½). Working: by Fleming's left-hand rule, forces on AB and CD are opposite, producing rotation (1). (b) Split ring reverses the current direction every half turn so the coil continues to rotate in the same direction (1½).